Reference 3-stage TEC-cooled HgCdTe device: independently reproducible physical and numerical method

Standalone scientific reproduction monograph | Model revision V3.3 | 8 November 2026 (UTC)

Scope and claim. This document completely states a realistic, engineering-plausible synthetic three-stage thermoelectric HgCdTe detector and package calculation. It does **not** represent a specific fabricated TEC Microsystems or VIGO product. All dimensions and laws are independently implementable without project history. The idealized pad geometry, measured-versus-assumed coefficients, and stage/circuit topology are explicitly identified. The model evaluates a complete 0.400 A current-driven electrothermal operating point and includes nonlinear temperature-dependent P/N transport, source-geometry ceramic 3D heat conduction, electrical solder Joule heat, detector load, hot mounting, Cu/Au links, and cavity gas/radiation loads.

Version distinction. A separate 38-page historical engineering assessment designated “Revision 3” primarily analyzed an earlier 216-solid 28/12/4 reduced configuration and a separate 3D solid-only 0.1 A case. This document instead describes the current 430-domain, 29/11/4 reference geometry and its coupled V3.3 working calculation. Never exchange numerical results between those differing geometries or boundary conditions.

1. Coordinate system, actual reference geometry and electrical topology

Coordinates use **millimetres** for mechanical geometry and **SI units** in equations. Set the package header normal as z. The four ceramic footprints and thicknesses are:

Ceramic	Side (mm)	Thickness (mm)	Top contact/thermal position
C0	6.4	0.500	C0 upper assembly pad
C1	5.2	0.500	C1 upper assembly pad
C2	3.8	0.500	C2 upper assembly pad
C3	2.4	0.500	C3 upper assembly pad

Source-composite design has **430 derived solid domains**, **2,440 native CAD face regions**, and **88 thermoelectric legs** connected in series, comprising **29 P/N couples in stage 0**, **11 in stage 1**, and **4 in stage 2**. The P and N legs are each 0.400 × 0.400 mm cross section and **0.500 mm length**, so individual $A = 1.600\times10^{-7}$ m² and $L = 5.000\times10^{-4}$ m. For electrical circuit orientation, individual current density magnitude is $J = 0.400\ A / A = 2.500\times10^{6}$ A m⁻², positive for the N leg and negative for the P leg under the source transport convention. The leg ordering, x/y centroids and polarities are tabulated in Appendix B, so the chosen virtual geometry can be regenerated independently.

There are two SnSb solder bodies per thermoelectric leg, **176 solder joints**, with **352** distinct source leg/solder and solder/metallization electrical interfaces. Every solder body has nominal thickness 35 µm and cross-sectional area 0.16 mm². Four original ceramic FEM regions use **11,791 linear four-node tetrahedra**; retained initial one-mesh region counts are 4,936 / 4,085 / 1,986 / 784 TET4. The finite-element mesh is a single saved discretization, not a demonstrated grid-converged thermal solution. Appendix C gives original ceramic-pad positions, areas and the final distributed loads.

A 1.0×1.0 mm² die is coupled to C3 by an assumed **20 µm EP29LPSP adhesive layer**, k = 1.0 W m⁻¹ K⁻¹; $R_{die\to carrier}$ = thickness/(kA) = **20.000 K W⁻¹**. No HgCdTe photocarrier PDE or optical transmission/absorption model is included.

2. Constitutive-law evidence and explicit model assumptions

Leg transport uses the published **J. Zhang et al., Chinese Physics Letters 42, 056304 (2025), DOI: 10.1088/0256-307X/42/5/056304**, figure-digitized data for distinct hot-extruded P-HE and N-HE-1 specimens. These are engineering analogs, not manufacturer-identified grades in this virtual design. Table 2A provides the entire 84-value, six-property input used. Numeric units: Seebeck **S** = μV K^{-1}, resistivity **rho** = μΩ m, and thermal conductivity **k** = W m^{-1} K^{-1}. Multiply S and rho columns by 10^{-6} to obtain SI values.

T (K)	S_P (μV/K)	rho_P (μΩ m)	k_P (W/mK)	S_N (μV/K)	rho_N (μΩ m)	k_N (W/mK)
40	31.477086	1.226120	2.776424	-39.762194	0.837035	3.928920
60	44.991486	1.550968	2.488935	-53.616397	1.156199	3.246955
80	60.679339	1.955036	2.260290	-67.469437	1.522950	2.816874
100	77.996604	2.446265	2.072406	-81.936140	1.920207	2.536709
120	96.944423	3.004830	1.909416	-95.992742	2.354830	2.310364
140	115.755946	3.567364	1.764537	-109.847998	2.820018	2.129254
160	133.684757	4.343818	1.630978	-122.547039	3.319204	1.973635
180	150.458980	5.175731	1.515526	-134.499967	3.835302	1.851971
200	166.078757	6.067058	1.415926	-145.840430	4.371811	1.744414
220	179.321421	7.073284	1.332165	-156.231255	4.932113	1.653884
240	190.594863	8.138932	1.257469	-165.737958	5.570441	1.574639
260	199.219885	9.351143	1.194083	-174.568277	6.232621	1.506729
280	205.739245	10.694094	1.137484	-182.445326	6.884545	1.447332
300	212.462550	12.080611	1.099006	-189.575244	7.543277	1.399227

Interpolate linearly between knots. The published range is **40–300 K**. In the modeled range **300–390 K**, the code uses a declared unmeasured continuation f(T) = f(300 K) + 0.25 (T−300 K) [f(300 K)−f(280 K)] / 20 K for f ∈ {S,rho,k}. All property evaluations must remain within **180–390 K**. Check positive rho and k and polarity S_P>0, S_N<0. Source-figure reading errors are not material characterization uncertainty. For V3.2, all 58 stage-0 legs required at least partial source-law extension above 300 K.

Ceramics: homogeneous hypothetical alumina k_c = **30 W m^{-1} K^{-1}**, constant from ~210 to 305 K; not an OEM measured k(T). All exposed ceramic faces not explicitly represented by load tags are insulated in the 3D conduction submodel.

3. Nonlinear one-dimensional thermoelectric conservation equations

For one leg, with z directed from hot to cold, current density J signed by polarity, temperature T(z), thermopower S(T), bulk electrical resistivity rho(T) and Fourier thermal conductivity k(T), use the conservative formulation:

$$q(z) = S(T)\,T\,J - k(T)\,dT/dz,$$

$$d\,q/dz = J\,[\rho(T)\,J + S(T)\,dT/dz],$$

$$dV/dz = \rho(T)\,J + S(T)\,dT/dz.$$

Using N=32 intervals, unknown $T_1 \ldots T_{31}$, fixed end temperatures $T_0=T_{hot}$, $T_{32}=T_{cold}$, and midpoint constitutive evaluations, define $q_i = S(T_{mid})\,T_{mid}\,J - k(T_{mid})(T_{i+1}-T_i)/\Delta z$, $\Delta V_i = J\,\rho(T_{mid})\,\Delta z + S(T_{mid})(T_{i+1}-T_i)$. The internal midpoint control-volume residual is $q_{i+1}-q_i - [J\,\Delta V_{i+1}+J\,\Delta V_i]/2 = 0$. Solve by damped Newton with a tridiagonal/banded Jacobian and positivity/domain checks. At boundaries include half-control-volume electrical work in q_hot and q_cold. Each leg must satisfy the integral conservation identity **$Q_{hot} - Q_{cold} - I\,V_{leg} = 0$**, using sign-consistent circuit voltage. The 29/11/4 stage heat and voltage are sums over the constituent P/N legs rather than a single fitted effective Seebeck coefficient.

4. Electrical solder, Cu and Au wire paths

For each SnSb joint: R_joint = rho_SnSb L_joint/A_joint + 2 r_c / A_joint. Use **rho_SnSb = 1.7×10⁻⁷ Ω m** and **specific electrical contact resistivity r_c = 1.0×10⁻¹¹ Ω m²**, hypothetical, not measured. For 176 joints in series this yields R_solder=0.006545 Ω, R_contact=0.022000 Ω, total added R=**0.028545 Ω**. At 0.400 A, direct voltage=**11.418 mV**, Joule heating=**4.56720 mW**. Place **25.950 µW** at each of 176 adjacent ceramic original metal pads; the resulting ceramic0..3 electrical joint heat vector is [1.50510,2.07600,0.77850,0.20760] mW. The all-Joule-into-ceramic partition is a provisional engineering simplification. It omits metal-net current crowding, finite interfacial thermal resistance and contact Peltier heat.

The two source Cu supply leads: radius 0.100 mm, lengths 3.451062 and 4.964887 mm, k=400 W/(m K), rho_e=1.72×10⁻⁸ Ωm, G=[0.0036413051,0.0025310487] W/K, R=[0.00188943245,0.00271824088] Ω. Model heat flow G_Cu (T_ambient−T_hot) and partition the total Cu I²R equally between room-temperature lead reservoir and hot ceramic. The two Au detector bonds: radius 9 µm, length 9.679142 mm each, k=315 W/(m K), G=8.28149215×10⁻⁶ W/K each. The Au electrical response is NOT used to calculate detector signal or photocurrent.

5. Three-dimensional ceramic steady heat conduction

For each ceramic C_i solve **−∇·(k_c∇T)=0** over its 3D source volume using continuous piecewise-linear tetrahedral FEM:

$$K_{ab} = \Sigma_t \int_{\Omega t} (\nabla N_a)\cdot k_c (\nabla N_b)\, d\Omega.$$

Contact heat loads are Neumann terms $b_a = \Sigma_f \int_{Af} N_a q_f dA$; for constant q_f per triangular face this assigns q_f A_f /3 to each of its three nodes. Native solder/metal pad heat is applied on its specific 3D TRI3 source surface. The source thermal operator is factorized once per ceramic; linear superposition updates the nonuniform surface heat distribution as the coupled leg flux changes. Temperature gauge is fixed by the lower pad area-mean temperature for that ceramic; no arbitrary artificial heat is introduced. Area-mean top minus bottom source temperatures couple to each stage hot/cold transport temperature; local individual leg temperatures are the proper native-pad mean, not all-leg averages.

For the present cavity prototype, the distributed physical gas and radiation heat incident on each cold exposed ceramic is **mapped to its upper native metal-pad basis in proportion to pad area**. This is an explicit reduced-source approximation, not a fully resolved bare-ceramic or gas-solid interface. A replica using continuous exposed-face Neumann flux may differ at local nodes.

6. Cavity gas and gray radiation: declared approximate physical closure

The cavity cap/window reservoir, hot mounting ambience and external lead reservoir are all held at **300.0 K**. The reference cold enclosure uses krypton at **100 Pa**, 3.00 mm effective gas gap, accommodation coefficient a=0.8, gas-area factor 1.0. At 200/300/400 K, NIST-derived Kr dilute-gas k values are [0.0064,0.0095,0.0122] W/(mK). Interpolate between tabulated points; the calculation uses the existing Eucken approximation $\mu = k/(c_p+1.25\,R_g)$, $c_p=\gamma R_g/(\gamma-1)$, γ=5/3, Rg=99.218 J/(kg K). The mean free path $\lambda = \mu/p \times \sqrt{\pi R_g T_{mean}/2}$. Thermal jump length $\delta = [(2-a)/a]\,[2\gamma/(\gamma+1)]\,\lambda/Pr$, $Pr=\mu c_p/k$. For p>0 the effective heat-transfer conductance is $G_{gas} = k A_{eff}/(L_{gap}+2\delta)$; $Q_{gas} = G_{gas}(300\,K - T_{cold})$. At p=0, Q_gas is defined as **exactly zero**. Mean temperatures below 200 K would invoke a separately flagged $k \propto T^{0.75}$ extension; the current case mean temperatures do not cross that limit. No buoyant convection, pressure-driven flow, CFD, molecular simulation, Xe mixture or free-convection model is claimed.

Effective cold-face areas from the source ceramic footprint differences (C1−C2, C2−C3, C3−die, die) are **[12.600,8.680,4.760,1.000] mm²**. For gray diffuse opaque exchange against a 300 K reservoir use $Q_{rad} = F\,\sigma\,A_{eff}\,\varepsilon_{eff}\,(300^4 - T_{cold}^4)$, $\varepsilon_{eff} = [1/\varepsilon_{warm} + 1/\varepsilon_{cold} - 1]^{-1}$, σ=5.670374419×10⁻⁸ W m⁻² K⁻⁴. Use ceramic ε_cold=0.80, cap ε_warm=0.20, cap view factor F=0.50 for three ceramic cold faces; use window ε_warm=0.10 and

window-to-die F=0.70 at the die. This is a gray effective-view reservoir approximation, not the 14-surface radiosity and chord network discussed in the historical Rev3. The cavity reservoir rejects the negative of the total gas plus radiation heat supplied to cold-stage and die nodes; record this separately to preserve thermodynamic energy accounting.

7. Full-device nonlinear coupling and boundary heat balances

Unknown temperatures are eight independent nodes: (T0h,T0c,T1h,T1c,T2h,T2c,Tcarrier,Tdie). The ceramic mean temperatures and local pad offsets are supplied by the four 3D FEM fields. Let Qh_i and Qc_i be total nonlinear P/N leg hot/cold heat for the three stages, Pj = 25.95 μW, and E1,E2,E3,Edie denote the local gas+radiation heat loads. The four conservative coupling equations are

$$Qc_0 - Qh_1 - (58+22)Pj - E1 = 0,$$

$$Qc_1 - Qh_2 - (22+8)Pj - E2 = 0,$$

$$Qc_2 - Qattach - 8Pj - E3 = 0,$$

$$Qattach - Pdie - QAu - Edie = 0.$$

Qattach = (Tdie−Tcarrier)/20 K W^{-1}, Pdie=0.002 W, QAu = ΣG_Au(300−Tdie). Source hot bath heat is Qbath=Qh_0+QCu+.5 I^2R_Cu+58Pj, QCu=ΣG_Cu(300−T0h). C0 bottom mean is **300 K + Rext Qbath** with Rext=**4.833386960805823 K/W** representing the assumed hot header/TIM/radiator path EXCLUDING the first 3D ceramic. Close the remaining four ceramic mean temperature-jump equations using their latest 3D P1 heat solution. For stage 0 hot side, T0h=300 K+Rext Qbath+ΔTceramic0.

The global steady electrical/thermal energy diagnostic is

$$\varepsilon_E = I\,V_{terminal} + Pdie + QAu + QCu - 0.5\,I^2R_{Cu}$$

$$+ E1+E2+E3+Edie - Qbath.$$

An accepted result has $|\varepsilon_E| < 10^{-8}$ W and each stage and source ceramic heat budget within its declared residual tolerance.

8. Solver algorithm that can be implemented without this source repository

1. Form source rectangular 29/11/4 P/N leg geometry, hot/cold

solder joints, four ceramic slabs, Cu/Au thermal leads and 1 mm^2 detector attach. Represent native pads using the appendix centroid/area table if a spatial reconstruction is desired.

2. Assemble each 3D ceramic linear-tetrahedron P1 conduction

stiffness K_i and factor once. Establish mean-pad-to-pad temperature influence vectors for the desired heat-source faces.

3. Initialize the eight temperatures from the fully specified

V3.2 enclosure-OFF reference values in Section 9.

4. Evaluate Qgas and Qrad at the latest cold ceramic and die

temperatures, then solve each of 88 P/N nonlinear axial 32-interval transport problems for its local hot/cold pad T.

5. Sum all leg Qhot/Qcold/V, add the 176 solder sources, Cu/Au

paths, detector 2 mW and cavity heat, and solve the eight-node nonlinear balance problem with bounded least-squares.

6. Recompute all four 3D ceramic fields with the new nonuniform

native pad Neumann heat sources and cavity loads. Repeat with outer under-relaxation 0.95 until maximum relative pad/plate temperature update is below **2×10^{-5} K**.

7. Reject any unphysical property, nonconverged stage, material

domain violation, incorrect source face-energy sum or global residual above $\mathbf{10^{-8}}$ **W**. Keep all numerical failure cases visible; no inferred parameters are adjusted to force convergence in this single registered reference case.

The outer stopping tolerance is intentionally **coarse** as an initial proof of coupled cavity physics. It is NOT a 20 µK independent mesh-convergence bound or as-built detector accuracy.

9. Independently checkable operating-point outputs

Case A is the identical complete model with cavity gas and radiation disabled (V3.2 source reference); **Case B** is the nominal 100 Pa Kr / gray radiation case (V3.3). All other dimensions, materials, current and external conditions are held fixed.

Quantity	Enclosure OFF (A)	Gas+radiation ON (B)
I (A)	0.4	0.4
Detector T (K)	209.9879814	210.625214
Terminal V (V)	1.524070832	1.522462416
Terminal P (W)	0.609628333	0.6089849664
Sink heat Qbath (W)	0.590322603	0.594043746
Global energy residual (W)	1.378857029e-10	8.867374612e-09
Ceramic TET4 count	11791	11791

Case B local enclosure inputs (mW):

Cold location	Gas input (mW)	Gray radiation input (mW)	Total (mW)
C1 exposed	1.15900974	0.20874186	1.36775160
C2 exposed	1.44533340	0.23389193	1.67922534
C3 exposed	1.06469312	0.15769702	1.22239014
Detector window/die	0.22351680	0.02374558	0.24726238
Total	**3.89255306**	**0.62407640**	**4.51662946**

Eight coupled temperature nodes in case B:

Node	T (K)
stage0_hot	303.65655327
stage0_cold	265.90328835
stage1_hot	266.34286716
stage1_cold	236.00258920
stage2_hot	236.15715707
stage2_cold	210.52764807
carrier	210.55066244
die	210.62521395

V3.3 versus V3.2 detector temperature: +0.63723256 K; **outer iterations:** 5; maximum final spatial update: 1.6535e-05 K.

Status: full 88-leg and four original ceramic P1 field coupled-device numerical calculation, not a fully spatial 430-domain electrical transport + rarefied-gas CFD + gray spectral radiation solver.

10. Verification and limitations before physical claims

- Full device energy identity: 8.8674e-09 W

- Maximum interstage/detector nodal mismatch: 4.9061e-09 W

- Maximum ceramic native-pad net energy: 5.5025e-15 W

- Maximum individual P/N leg energy residual: 1.2364e-13 W

- Final spatial temperature update: 1.6535e-05 K

- The historical original P1 whole-assembly electrical Gate B remained FAIL/STOP;

the first source P/N pair raw gradient-current test failed 8/8, prospective P2 stopped, and the original full 430-domain coupled 3D C/D thermal qualification has NOT RUN.

- No OEM grade, installed cooler assembly or detector temperature

measurement is claimed; neither numerical residual nor plotted precision is a manufacturing accuracy bound.

- Not modeled: true 3D P/N current transport, temperature-dependent

ceramic grade, solder interfacial thermopower, cavity gas CFD, all-surface radiosity, optical window transmission, semiconductor carrier transport, thermistor or transient control.

Appendix B. Complete original-source 88-leg placement for independent replication

Leg area is 0.1600 mm², length 0.5000 mm; coordinates are CAD-body bounding-box centre x/y in mm. Stage0 has 58 legs, stage1 22, stage2 8.

Series index	Stage	P/N	X-centre (mm)	Y-centre (mm)
0	0	N	-2.030000	-2.030000
1	0	P	-1.450000	-2.030000
2	0	N	-0.870000	-2.030000
3	0	P	-0.290000	-2.030000
4	0	N	0.290000	-2.030000
5	0	P	0.870000	-2.030000
6	0	N	1.450000	-2.030000
7	0	P	2.030000	-2.030000
8	0	N	2.030000	-1.450000
9	0	P	1.450000	-1.450000
10	0	N	0.870000	-1.450000
11	0	P	0.290000	-1.450000
12	0	N	-0.290000	-1.450000
13	0	P	-0.870000	-1.450000
14	0	N	-1.450000	-1.450000
15	0	P	-2.030000	-1.450000
16	0	N	-2.030000	-0.870000
17	0	P	-1.450000	-0.870000
18	0	N	-0.870000	-0.870000
19	0	P	-0.290000	-0.870000

Series index	Stage	P/N	X-centre (mm)	Y-centre (mm)
20	0	N	0.290000	-0.870000
21	0	P	0.870000	-0.870000
22	0	N	1.450000	-0.870000
23	0	P	2.030000	-0.870000
24	0	N	2.030000	-0.290000
25	0	P	1.450000	-0.290000
26	0	N	0.870000	-0.290000
27	0	P	0.290000	-0.290000
28	0	N	-0.290000	-0.290000
29	0	P	-0.870000	-0.290000
30	0	N	-1.450000	-0.290000
31	0	P	-2.030000	-0.290000
32	0	N	-2.030000	0.290000
33	0	P	-1.450000	0.290000
34	0	N	-0.870000	0.290000
35	0	P	-0.290000	0.290000
36	0	N	0.290000	0.290000
37	0	P	0.870000	0.290000
38	0	N	1.450000	0.290000
39	0	P	2.030000	0.290000
40	0	N	2.030000	0.870000
41	0	P	1.450000	0.870000
42	0	N	0.870000	0.870000
43	0	P	0.290000	0.870000
44	0	N	-0.290000	0.870000
45	0	P	-0.870000	0.870000
46	0	N	-1.450000	0.870000
47	0	P	-2.030000	0.870000
48	0	N	-2.030000	1.450000
49	0	P	-1.450000	1.450000
50	0	N	-0.870000	1.450000
51	0	P	-0.290000	1.450000
52	0	N	0.290000	1.450000
53	0	P	0.870000	1.450000
54	0	N	1.450000	1.450000
55	0	P	2.030000	1.450000
56	0	N	2.030000	2.030000
57	0	P	1.450000	2.030000
58	1	N	1.450000	0.870000
59	1	P	0.870000	0.870000

Series index	Stage	P/N	X-centre (mm)	Y-centre (mm)
60	1	N	0.290000	0.870000
61	1	P	-0.290000	0.870000
62	1	N	-0.870000	0.870000
63	1	P	-1.450000	0.870000
64	1	N	-1.450000	0.290000
65	1	P	-0.870000	0.290000
66	1	N	-0.290000	0.290000
67	1	P	0.290000	0.290000
68	1	N	0.870000	0.290000
69	1	P	1.450000	0.290000
70	1	N	1.450000	-0.290000
71	1	P	0.870000	-0.290000
72	1	N	0.290000	-0.290000
73	1	P	-0.290000	-0.290000
74	1	N	-0.870000	-0.290000
75	1	P	-1.450000	-0.290000
76	1	N	-1.450000	-0.870000
77	1	P	-0.870000	-0.870000
78	1	N	-0.290000	-0.870000
79	1	P	0.290000	-0.870000
80	2	N	0.870000	-0.290000
81	2	P	0.290000	-0.290000
82	2	N	-0.290000	-0.290000
83	2	P	-0.870000	-0.290000
84	2	N	-0.870000	0.290000
85	2	P	-0.290000	0.290000
86	2	N	0.290000	0.290000
87	2	P	0.870000	0.290000

Appendix C. Native ceramic-pad geometry and solved surface heat flux

The following per-face heat source table (W) includes the true individual leg heat and solder/contact Joule, plus the separately declared area-weighted gas/radiation surrogate and hot Cu/sink/die terms. Face centroids and exact CAD areas are given for an independent geometrical placement; a matching custom mesh will not reproduce exact local gradients without the complete original face boundary polygon outlines.

Ceramic	Face ID	X (mm)	Y (mm)	Z (mm)	CAD area (mm²)	Net heat into ceramic (W)
0	6	0.000000	0.000000	1.584250	37.21000000	-5.9404375264e-01
0	8	-2.030000	-2.415000	2.084250	0.58100000	+6.9212449542e-03
0	9	-1.160000	-2.030000	2.084250	0.54000000	+2.1001118011e-02
0	10	-0.000000	-2.030000	2.084250	0.54000000	+2.0799385661e-02
0	11	-2.030000	-1.160000	2.084250	0.54000000	+2.1333430585e-02

Ceramic	Face ID	X (mm)	Y (mm)	Z (mm)	CAD area (mm²)	Net heat into ceramic (W)
0	12	-1.160000	-1.450000	2.084250	0.54000000	+2.0925106301e-02
0	13	-1.160000	-0.870000	2.084250	0.54000000	+2.1098181465e-02
0	14	-2.030000	-0.000000	2.084250	0.54000000	+2.1661444954e-02
0	15	-1.160000	-0.290000	2.084250	0.54000000	+2.1280527762e-02
0	16	-0.000000	-1.450000	2.084250	0.54000000	+2.0726535211e-02
0	17	0.000000	-0.870000	2.084250	0.54000000	+2.0872339885e-02
0	18	-0.000000	-0.290000	2.084250	0.54000000	+2.1135046258e-02
0	19	1.160000	-2.030000	2.084250	0.54000000	+2.0762274895e-02
0	20	2.030000	-1.740000	2.084250	0.54000000	+2.0894962316e-02
0	21	1.160000	-1.450000	2.084250	0.54000000	+2.0664089207e-02
0	22	1.160000	-0.870000	2.084250	0.54000000	+2.0730610371e-02
0	23	1.160000	-0.290000	2.084250	0.54000000	+2.0987932670e-02
0	24	2.030000	-0.580000	2.084250	0.54000000	+2.1025501033e-02
0	25	-1.160000	0.290000	2.084250	0.54000000	+2.1501510390e-02
0	26	-2.030000	1.160000	2.084250	0.54000000	+2.2056807823e-02
0	27	-1.160000	0.870000	2.084250	0.54000000	+2.1665224003e-02
0	28	-1.160000	1.450000	2.084250	0.54000000	+2.1972015528e-02
0	29	0.000000	0.290000	2.084250	0.54000000	+2.1329922044e-02
0	30	-0.000000	0.870000	2.084250	0.54000000	+2.1513476612e-02
0	31	0.000000	1.450000	2.084250	0.54000000	+2.1762448001e-02
0	32	1.160000	0.290000	2.084250	0.54000000	+2.1132384800e-02
0	33	2.030000	0.580000	2.084250	0.54000000	+2.1315494607e-02
0	34	1.160000	0.870000	2.084250	0.54000000	+2.1316483722e-02
0	35	1.160000	1.450000	2.084250	0.54000000	+2.1497446937e-02
0	36	0.088570	2.858995	2.084250	0.67225000	-2.2200926469e-02
0	37	2.030000	1.740000	2.084250	0.54000000	+2.1652376653e-02
0	38	1.450000	2.379622	2.084250	0.51850000	+1.4709356454e-02
1	44	-1.740000	-2.030000	2.684250	0.54000000	-6.1717204457e-03
1	45	-0.580000	-2.030000	2.684250	0.54000000	-5.7923394592e-03
1	46	0.580000	-2.030000	2.684250	0.54000000	-5.6313481447e-03
1	47	-1.740000	-1.450000	2.684250	0.54000000	-5.8945158835e-03
1	48	-1.740000	-0.870000	2.684250	0.54000000	-6.1272498337e-03
1	49	-0.580000	-1.450000	2.684250	0.54000000	-5.6677985879e-03
1	50	-0.580000	-0.870000	2.684250	0.54000000	-5.9073535857e-03
1	51	-1.740000	-0.290000	2.684250	0.54000000	-6.2861601984e-03
1	52	-1.740000	0.290000	2.684250	0.54000000	-6.4929448577e-03
1	53	-0.580000	-0.290000	2.684250	0.54000000	-6.1441766581e-03
1	54	0.580000	-1.450000	2.684250	0.54000000	-5.4949366062e-03
1	55	0.580000	-0.870000	2.684250	0.54000000	-5.6165637076e-03
1	56	0.580000	-0.290000	2.684250	0.54000000	-5.9284010470e-03

Ceramic	Face ID	X (mm)	Y (mm)	Z (mm)	CAD area (mm²)	Net heat into ceramic (W)
1	57	1.740000	-2.030000	2.684250	0.54000000	-5.5964450493e-03
1	58	1.740000	-1.450000	2.684250	0.54000000	-5.5750930796e-03
1	59	1.740000	-0.870000	2.684250	0.54000000	-5.5316729855e-03
1	60	1.740000	-0.290000	2.684250	0.54000000	-5.7862099653e-03
1	61	-0.580000	0.290000	2.684250	0.54000000	-6.4257386885e-03
1	62	-1.740000	0.870000	2.684250	0.54000000	-6.8576261766e-03
1	63	-1.740000	1.450000	2.684250	0.54000000	-7.0890067759e-03
1	64	-0.580000	0.870000	2.684250	0.54000000	-6.6249732312e-03
1	65	-0.580000	1.450000	2.684250	0.54000000	-7.0137372522e-03
1	66	0.580000	0.290000	2.684250	0.54000000	-6.1746390641e-03
1	67	0.580000	0.870000	2.684250	0.54000000	-6.3948651910e-03
1	68	0.580000	1.450000	2.684250	0.54000000	-6.7126784447e-03
1	69	1.740000	0.290000	2.684250	0.54000000	-6.0016216656e-03
1	70	1.740000	0.870000	2.684250	0.54000000	-6.1198269805e-03
1	71	1.740000	1.450000	2.684250	0.54000000	-6.5247521871e-03
1	72	1.740000	2.030000	2.684250	0.54000000	-6.8707555753e-03
1	74	0.870000	2.300000	3.184250	0.12000000	+2.3941388903e-05
1	75	1.450000	1.470862	3.184250	0.67900000	+5.1504788556e-03
1	76	-1.450000	-0.580000	3.184250	0.54000000	+1.6210670281e-02
1	77	-0.580000	-0.870000	3.184250	0.54000000	+1.6207954022e-02
1	78	-0.580000	-0.290000	3.184250	0.54000000	+1.6071664242e-02
1	79	0.290000	-1.435162	3.184250	0.65650000	+1.1539642240e-02
1	80	0.580000	-0.290000	3.184250	0.54000000	+1.6217492734e-02
1	81	1.450000	0.000000	3.184250	0.54000000	+1.6376311378e-02
1	82	-1.450000	0.580000	3.184250	0.54000000	+1.6153119097e-02
1	83	-0.580000	0.290000	3.184250	0.54000000	+1.6048683270e-02
1	84	-0.580000	0.870000	3.184250	0.54000000	+1.6098735692e-02
1	85	0.580000	0.290000	3.184250	0.54000000	+1.6138044094e-02
1	86	0.580000	0.870000	3.184250	0.54000000	+1.6218414034e-02
2	92	-1.160000	-0.870000	3.784250	0.54000000	-4.1229763184e-03
2	93	-1.160000	-0.290000	3.784250	0.54000000	-4.0325885849e-03
2	94	0.000000	-0.870000	3.784250	0.54000000	-4.3372223253e-03
2	95	-0.000000	-0.290000	3.784250	0.54000000	-4.1255544021e-03
2	96	1.160000	-0.290000	3.784250	0.54000000	-4.2477066490e-03
2	97	-1.160000	0.290000	3.784250	0.54000000	-3.9991911145e-03
2	98	-1.160000	0.870000	3.784250	0.54000000	-4.0132490723e-03
2	99	0.000000	0.290000	3.784250	0.54000000	-4.0452781812e-03
2	100	1.160000	0.290000	3.784250	0.54000000	-4.1617896577e-03
2	101	-0.000000	0.870000	3.784250	0.54000000	-4.1215621922e-03
2	102	1.160000	0.870000	3.784250	0.54000000	-4.3855401752e-03

Ceramic	Face ID	X (mm)	Y (mm)	Z (mm)	CAD area (mm²)	Net heat into ceramic (W)
2	104	0.870000	0.829062	4.284250	0.64000000	+8.5720985048e-03
2	105	0.692720	-0.929413	4.284250	0.81350000	+3.2580848072e-03
2	106	-0.000000	-0.290000	4.284250	0.54000000	+1.1238504436e-02
2	107	-0.870000	0.000000	4.284250	0.54000000	+1.1290111066e-02
2	108	0.000000	0.290000	4.284250	0.54000000	+1.1233859859e-02
3	114	0.580000	0.290000	4.884250	0.54000000	-1.2327805666e-03
3	115	-0.580000	-0.290000	4.884250	0.54000000	-1.2229792254e-03
3	116	0.580000	-0.290000	4.884250	0.54000000	-1.2662612022e-03
3	117	-0.580000	0.290000	4.884250	0.54000000	-1.2279447029e-03
3	119	-0.000000	-0.000000	5.384250	4.41000000	+4.9499656971e-03

Independent references and source limitations

1. J. Zhang et al., “Accurate Measurements and Error Analysis of

Bi2Te3-Based Low-Temperature Thermoelectrics,” Chinese Physics Letters 42, 056304 (2025). DOI: 10.1088/0256-307X/42/5/056304. Six curve values here were digitized from published plots, not a commercial TEC factory material certificate.

2. M. L. Huber, “Thermal Conductivity of Gases,” NIST/CRC (2019).

Reference Kr dilute-gas thermal-conductivity values at 200–400 K.

3. E. M. Heckman, L. P. Gonzalez, S. Guha, “Measurement of optical

and thermal properties of Hg1−xCdxTe,” Applied Optics 47, 578–582 (2008). DOI: 10.1364/AO.47.000578.

4. TEC Microsystems 3MC04-044-05 datasheet (2019), independent

commercial performance envelope, not proof that the present synthetic TEC assembly or P/N materials are an identical product.

5. VIGO Photonics 3TE-TO66 9-pin reference package drawing, Rev. 9

(2024); family-scale dimensional reference, not as-built validation.

Replication policy. Reported output numerics are conditional simulation results. A fully independent implementation can reproduce physically similar stage and detector responses with this geometry, material table, equations, heat paths and inputs. Bitwise field equivalence requires identical unstructured mesh, per-face polygons, quadrature, solver libraries and tolerances; those are not implied by this standalone report.